

Q1. Classification accuracy is defined as $\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$. Its main limitation is that it assumes equal importance for all classes and equal misclassification cost. In an imbalanced dataset with 99% negatives and 1% positives, a classifier predicting every sample as negative yields 99% accuracy while completely failing to detect positives. This is called the accuracy paradox: numerically high accuracy but practically useless prediction. Theoretically, accuracy is dominated by majority-class prior probability $P(Y=\text{majority})$. For skewed priors, the Bayes trivial classifier may maximize accuracy while minimizing utility. Alternative metrics include precision $TP / (TP + FP)$, recall $TP / (TP + FN)$, F1-score $2PR / (P + R)$, ROC-AUC, PR-AUC, balanced accuracy $= (\text{sensitivity} + \text{specificity}) / 2$, and Matthews correlation coefficient. In fraud detection or rare disease prediction, recall and PR-AUC are preferred because false negatives carry larger practical cost.

Q2. Precision $= TP / (TP + FP)$, Recall $= TP / (TP + FN)$, Specificity $= TN / (TN + FP)$. F1-score is the harmonic mean: $F1 = 2 \cdot (\text{Precision} \cdot \text{Recall}) / (\text{Precision} + \text{Recall})$. Precision emphasizes reliability of positive predictions, while recall emphasizes completeness of positive detection. Under extreme imbalance, specificity often remains high because TN dominates, even when recall is poor. For example, if $TP=5$, $FN=95$, $FP=5$, $TN=9995$, recall $= 0.05$ while specificity ≈ 0.9995 . F1 becomes sensitive because low recall strongly reduces harmonic mean. In medical diagnosis, high recall is prioritized to avoid missed disease cases, but precision also matters to reduce unnecessary interventions. Hence threshold tuning is often done to maximize recall under acceptable precision.

Q3. ROC curve plots True Positive Rate (TPR) against False Positive Rate (FPR) for varying thresholds τ on predicted score $s(x)$. $TPR(\tau) = P(s(x) \geq \tau | Y=1)$, $FPR(\tau) = P(s(x) \geq \tau | Y=0)$. Lowering τ classifies more samples as positive, increasing both TPR and FPR monotonically. Probabilistically, ROC represents separability between score distributions $f_1(s)$ and $f_0(s)$. A perfect classifier has curve through (0,1) and $AUC=1$; random guessing gives diagonal $AUC=0.5$. The derivative $dTPR/dFPR$ equals likelihood ratio $f_1(\tau)/f_0(\tau)$. Optimal threshold is often chosen near maximum Youden index $J = TPR - FPR$. ROC helps compare classifiers independent of class prior and is robust when operating thresholds differ across deployment conditions.

Q4. Parametric models assume a fixed functional form with finite parameters, such as linear regression or logistic regression. Non-parametric models such as k-NN or decision trees allow complexity to grow with data size. Parametric models generally have higher bias and lower variance because hypothesis space is restricted. Non-parametric models reduce bias but may increase variance due to flexibility. Parametric methods scale efficiently because inference depends on parameter dimension rather than dataset size. Non-parametric methods often require storing training data and can have slower inference. Parametric models assume distributions or structural relations; for example logistic regression assumes log-odds linearity. Non-parametric models adapt better to irregular distributions. (Intentional mistake:) k-NN is always computationally cheaper than logistic regression for very large datasets.

Q5. Overfitting occurs when a model captures noise in training data, producing low training error but high test error. Underfitting occurs when model capacity is insufficient, leading to high error on both training and test sets. Bias-variance decomposition expresses expected error as $\text{Bias}^2 + \text{Variance} + \text{irreducible noise}$. High bias causes underfitting; high variance causes overfitting. Cross-validation estimates generalization by repeatedly partitioning data into training and validation folds. In k-fold cross-validation, each fold serves once as validation and $k-1$ times as training, reducing dependence on a single split. It helps select hyperparameters such as depth, regularization strength, or kernel width that minimize expected validation error, thereby improving generalization.